

Exam 2 - Practice

Instructions:

1. You have 50 min to complete this exam.
2. Write all the process you did to solve a problem.
3. You can use your notes, but take care of your time if you do it.

Problem 1. Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}, \quad \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

- (i) Show that $\mathcal{B}$ is a basis of $\mathbb{R}^2$. Justify.
- (ii) Compute the matrix that change basis from $\mathcal{B}$ to $\mathcal{C}$ and from $\mathcal{C}$ to $\mathcal{B}$ ($[I]_{\mathcal{C}}^{\mathcal{B}}$ and $[I]_{\mathcal{B}}^{\mathcal{C}}$ respectively), where $\mathcal{C}$ is the canonical basis.
- (iii) Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. What are the coordinates of $\mathbf{v}$ in the canonical basis $\mathcal{C}$? And what are the coordinates of $\mathbf{v}$ in the basis $\mathcal{B}$? Justify.

Problem 2. Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{D} = \{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_4\}$ be bases for vector spaces V and W , respectively. Let $T: V \rightarrow W$ be a linear transformation with the property that

$$T(\mathbf{b}_1) = 3\mathbf{d}_1 - 5\mathbf{d}_2 + 9\mathbf{d}_3 \quad T(\mathbf{b}_2) = -\mathbf{d}_1 + 6\mathbf{d}_2 - \mathbf{d}_4,$$

Find the matrix for T relative to $\mathcal{B}$ and $\mathcal{D}$. That is, compute $[T]_{\mathcal{D}}^{\mathcal{B}}$.

Problem 3. Compute the characteristic polynomial of the following matrix.

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

Problem 4. Let A be the following matrix.

$$A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

The characteristic polynomial of A is $p(\lambda) = (1 - \lambda)^3(2 - \lambda)$

- (i) What are the eigenvalues of A ? Justify.
- (ii) What are the algebraic multiplicities of the eigenvalues of A ?
- (iii) Compute a basis for the eigenspace corresponding to the smallest eigenvalue.
- (iv) What are the geometric multiplicities of all the eigenvalues? Justify.
- (v) Is A diagonalizable? why?

Problem 5. Let A be the following matrix

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

The eigenvalues of A are $\lambda = 2, 4$ and the eigenvectors are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ respectively. Find a matrix S such that $A = SDS^{-1}$ where D is a diagonal matrix. You might use the results of problem 1.

Bonus Problem. Let W be the subspace of $\mathbb{R}^4$ generated by the vectors

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Compute a basis for the orthogonal complement subspace $W^\perp$ of W .